Matrix Factorization via Poisson SuSiE

1 Model

We observe a count matrix $Y \in \mathbb{N}^{N\times M}$. Each row y_i is modeled as

$$y_{im} \sim \mathrm{Poi}\Big(\sum_{d=1}^{D} \beta_{id}\, F_{\gamma_{id},\, m}\Big), \quad m \in [M],$$

where $F \in \mathbb{R}_+^{K\times M}$ is a shared factor matrix satisfying $\sum_m F_{km} = 1$ for all k, and for each observation i and effect d:

$$\gamma_{id} \sim \mathrm{Multi}(1,\, \tfrac{1}{K}\mathbf{1}_K), \tag{1}$$

$$\beta_{id} \mid \gamma_{id} \sim \mathrm{Gam}(\alpha^0_{id},\, \lambda^0_{id}). \tag{2}$$

2 Variational family

We use the mean-field variational family

$$q(\beta, \gamma) = \prod_{i=1}^{N}\prod_{d=1}^{D} q(\gamma_{id})\, q(\beta_{id} \mid \gamma_{id}),$$

where $q(\gamma_{id}) = \mathrm{Multi}(1,\, \bar{\gamma}_{id})$ with $\bar{\gamma}_{id} = (\bar{\gamma}_{id1}, \ldots, \bar{\gamma}_{idK})$, and $q(\beta_{id} \mid \gamma_{id} = k) = \mathrm{Gam}(\alpha_{idk},\, \lambda_{idk})$.

We also introduce auxiliary variables $\xi_{imd} \geq 0$ with $\sum_d \xi_{imd} = 1$ for each (i, m).

3 Variational parameters

The optimal variational parameters are:

$$\alpha_{idk} = \sum_m \xi_{imd}\, y_{im} + \alpha^0_{id}, \tag{3}$$

$$\lambda_{idk} = \sum_m F_{km} + \lambda^0_{id}. \tag{4}$$

Note that α_{idk} does not depend on k, and λ_{idk} does not depend on i.

3.1 Moments

Under $q(\beta_{id} \mid \gamma_{id} = k) = \mathrm{Gam}(\alpha_{idk}, \lambda_{idk})$:

$$\mathbb{E}_{q(\beta_{id}|\gamma_{id}=k)}[\log \beta_{id}] = \psi(\alpha_{idk}) - \log(\lambda_{idk}), \tag{5}$$

$$\mathbb{E}_{q(\beta_{id},\, \gamma_{id})}[\log \beta_{id}] = \sum_k \bar{\gamma}_{idk}\big[\psi(\alpha_{idk}) - \log(\lambda_{idk})\big]. \tag{6}$$

4 ELBO

The evidence lower bound is

$$\mathcal{E}(q) = \sum_{i=1}^{N}\sum_{d=1}^{D} \mathbb{E}_{q_{id}(\gamma_{id},\,\beta_{id})}\Big[\sum_m \big(\xi_{imd}\, y_{im}(\log \beta_{id} + \log F_{\gamma_{id},\,m}) - \beta_{id}\, F_{\gamma_{id},\,m}\big)\Big]$$
$$- \sum_{i=1}^{N}\sum_{d=1}^{D} \mathrm{KL}\big(q(\gamma_{id},\,\beta_{id})\,\|\,p(\gamma_{id},\,\beta_{id})\big) - \sum_{i,m,d} y_{im}\,\xi_{imd}\,\log \xi_{imd}. \tag{7}$$

Expanding term by term:

$$\text{Term 1} = \sum_{i,d,m} \xi_{imd}\, y_{im}\, \mathbb{E}_q[\log \beta_{id}], \tag{8}$$

$$\text{Term 2} = \sum_{i,d,m} \xi_{imd}\, y_{im} \sum_k \bar{\gamma}_{idk}\, \log F_{km}, \tag{9}$$

$$\text{Term 3} = -\sum_{i,d,m}\sum_k \bar{\gamma}_{idk}\, \frac{\alpha_{idk}}{\lambda_{idk}}\, F_{km}, \tag{10}$$

$$\text{Term 4} = -\sum_{i,m,d} y_{im}\,\xi_{imd}\,\log \xi_{imd}. \tag{11}$$

The KL divergence decomposes as $\mathrm{KL} = \mathrm{KL}_{\text{disc}} + \mathrm{KL}_{\text{cont}}$:

$$\mathrm{KL}_{\text{disc}} = \sum_{i,d,k} \bar{\gamma}_{idk}\big(\log \bar{\gamma}_{idk} - \log \tfrac{1}{K}\big), \tag{12}$$

$$\mathrm{KL}_{\text{cont}} = \sum_{i,d,k} \bar{\gamma}_{idk}\Big[(\alpha_{idk} - \alpha^0_{id})\,\psi(\alpha_{idk}) + (\log \lambda_{idk} - \log \lambda^0_{id})\,\alpha^0_{id}$$
$$- \log \frac{\Gamma(\alpha_{idk})}{\Gamma(\alpha^0_{id})} - (\lambda_{idk} - \lambda^0_{id})\,\frac{\alpha_{idk}}{\lambda_{idk}}\Big]. \tag{13}$$

5 Algorithm

The current implementation uses several stabilization and speed options. When F is unknown, it can first run a standard Poisson/KL-NMF fit $Y \approx LF$ with row-normalized F and use the resulting F as the initial shared factor matrix. The corresponding NMF loading estimates can also initialize $\bar{\gamma}_{id}$ by assigning the D effects of observation i to the largest entries of the NMF loading row. During the MF-Poisson-SuSiE iterations, the categorical update can be damped with step size ρ_t, and the F update can be damped with step size η_t. The default algorithm refreshes the auxiliary responsibilities ξ before each effect update, which is a Gauss-Seidel-style schedule. The ELBO may be evaluated only every few iterations because it is not needed for the coordinate updates.

Algorithm 1 MF-Poisson-SuSiE with learned F

Initialize F by either row-normalizing the supplied F or by running Poisson/KL-NMF and row-normalizing the resulting factor rows.
Initialize $\bar{\gamma}$ either randomly or from the top entries of the Poisson-NMF loading rows.
Initialize α, λ, and $\xi_{imd} = 1/D$.
for iter $= 1, \dots, T$ **do**
  Set damping weights ρ_t for $\bar{\gamma}$ and η_t for F.
  (Break symmetry) Merge duplicate effects if needed.
  for $d = 1, \dots, D$ **do**
    E-step: Refresh ξ using all current effects:

$$\xi_{imd} \leftarrow \text{Norm-Exp}_d\Big(\mathbb{E}_{q(\gamma_{id},\, \beta_{id})}[\log \beta_{id} + \log F_{\gamma_{id},\, m}]\Big), \quad \forall\, i \in [N],\, m \in [M].$$

    M-step: Update variational parameters for effect d:

$$\alpha_{idk} \leftarrow \sum_m \xi_{imd}\, y_{im} + \alpha^0_{id},$$
$$\lambda_{idk} \leftarrow \sum_m F_{km} + \lambda^0_{id}.$$

    PIP: Compute logprob_{idk} and update $\bar{\gamma}_{id}$:

$$\text{logprob}_{idk} = \sum_m \xi_{imd}\, y_{im}\, \log F_{km} - \alpha_{idk}\, \log \lambda_{idk},$$
$$\bar{\gamma}^{\text{new}}_{id} \leftarrow \text{Norm-Exp}(\text{logprob}_{id1}, \dots, \text{logprob}_{idK}),$$
$$\bar{\gamma}_{id} \leftarrow (1 - \rho_t)\bar{\gamma}_{id} + \rho_t \bar{\gamma}^{\text{new}}_{id}.$$

    VEB (optional): Update $(\alpha^0_{id}, \lambda^0_{id})$ using the empirical-Bayes equations in Section 7.
  end for
  Refresh ξ using the final $\bar{\gamma}$, α, and λ from this outer iteration.
  Update F (optional): For each k, m:

$$A_{km} = \sum_{i,d} \bar{\gamma}_{idk}\, \xi_{imd}\, y_{im},$$
$$F^{\text{new}}_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'}(A_{km'} + \epsilon)},$$
$$F_{km} \leftarrow (1 - \eta_t)F_{km} + \eta_t F^{\text{new}}_{km}, \qquad F_{k\cdot} \leftarrow \frac{F_{k\cdot}}{\sum_{m'} F_{km'}}.$$

  Refresh ξ after updating F.
  if ELBO evaluation is scheduled for this iteration **then**
    Evaluate $\mathcal{E}(q)$.
  end if
end for

Remark 1 (Alternative ξ refresh schedules)**.** Cheaper variants of the ξ refresh are possible. A Jacobi-style schedule refreshes ξ once per outer iteration and reuses it for all D effect updates. A hybrid schedule uses the Gauss-Seidel refresh for an initial warm-up period, then switches to the Jacobi-style refresh. These variants do not change the model or the coordinate updates; they only trade off local optimization behavior against speed.

6 Derivation of the F update

The terms in the ELBO involving F_{km} are:

$$\sum_{k,m} \Big[A_{km} \, \log F_{km} - B_k \, F_{km} \Big],$$

where

$$A_{km} = \sum_{i,d} \bar{\gamma}_{idk} \, \xi_{imd} \, y_{im}, \qquad B_k = \sum_{i,d} \bar{\gamma}_{idk} \, \frac{\alpha_{idk}}{\lambda_{idk}}.$$

Maximizing subject to $\sum_m F_{km} = 1$ via Lagrange multipliers:

$$\frac{\partial}{\partial F_{km}} \Big[A_{km} \, \log F_{km} - B_k \, F_{km} - \mu_k \big(\sum_m F_{km} - 1 \big) \Big] = 0 \implies F_{km} = \frac{A_{km}}{B_k + \mu_k}.$$

Applying the constraint: $B_k + \mu_k = \sum_m A_{km}$, hence

$$F_{km} = \frac{A_{km}}{\sum_{m'} A_{km'}}.$$

7 Derivation of the VEB updates

The expected log-prior for effect d of observation i is:

$$\mathbb{E}_q[\log p(\beta_{id} \mid \gamma_{id};\, \alpha^0_{id},\, \lambda^0_{id})] = \sum_k \bar{\gamma}_{idk} \Big[\alpha^0_{id} \log \lambda^0_{id} - \log \Gamma(\alpha^0_{id}) + (\alpha^0_{id} - 1) \big(\psi(\alpha_{idk}) - \log \lambda_{idk} \big) - \lambda^0_{id} \, \frac{\alpha_{idk}}{\lambda_{idk}} \Big].$$

λ^0_{id} update (closed form). Setting $\frac{\partial}{\partial \lambda^0_{id}} = 0$:

$$\sum_k \bar{\gamma}_{idk} \Big[\frac{\alpha^0_{id}}{\lambda^0_{id}} - \frac{\alpha_{idk}}{\lambda_{idk}} \Big] = 0 \implies \lambda^0_{id} = \frac{\alpha^0_{id}}{\sum_k \bar{\gamma}_{idk} \, \alpha_{idk} / \lambda_{idk}} = \frac{\alpha^0_{id}}{\bar{\beta}_{id}}.$$

α^0_{id} update (implicit equation). Setting $\frac{\partial}{\partial \alpha^0_{id}} = 0$:

$$\sum_k \bar{\gamma}_{idk} \Big[\log \lambda^0_{id} - \psi(\alpha^0_{id}) + \psi(\alpha_{idk}) - \log \lambda_{idk} \Big] = 0 \implies \psi(\alpha^0_{id}) = \overline{\log \beta}_{id} + \log \lambda^0_{id}.$$

This is solved by inverting the digamma function, e.g., via Newton's method:

$$\alpha^0_{id} \leftarrow \alpha^0_{id} - \frac{\psi(\alpha^0_{id}) - \text{target}}{\psi'(\alpha^0_{id})}, \qquad \text{target} = \overline{\log \beta}_{id} + \log \lambda^0_{id}.$$